🚀 SECTION 7 — PROMPT ENGINEERING MASTERY

🎓 Complete Learner System (Use While Watching)

🧭 HOW TO USE THIS SECTION

For each lesson:

1. Watch
2. Rewrite the prompt yourself
3. Test it immediately
4. Improve it
5. Save your best version

👉 This section is about **practice + iteration**

🧠 SECTION 7 — CORE IDEA

Better input = better output
Prompting is not asking — it is **designing instructions**

📘 PART 1 — GUIDED NOTES

📘 L7_01 — What Is Prompt Engineering

Key Ideas:

- Prompt = instruction system
- AI follows input patterns
- Quality of input = output quality

Write:

- What makes a prompt “good”?

📘 L7_02 — Anatomy of a Prompt

Key Ideas:

- Instruction
- Context
- Output format

Write:

- Which part do you usually miss?

📘 L7_03 — Prompt Templates

Key Ideas:

- Reusable prompts
- Consistency
- Speed

Write:

- What do you repeat often?

📘 L7_04 — Iterative Prompting

Key Ideas:

- First output is not final
- Improve step-by-step

Write:

- Do you refine or accept?

📘 L7_05 — Advanced Prompting

Key Ideas:

- Structured input
- Constraints
- Multi-layer prompts

Write:

- How can you add structure?

📘 L7_06 — Role Prompting

Key Ideas:

- Assign perspective
- Changes tone + depth

Write:

- What role would improve results?

📘 L7_07 — Step-by-Step Prompting

Key Ideas:

- Break tasks into steps

- Improves clarity

Write:

- What task can you break down?

📘 L7_08 — Constraint Prompting

Key Ideas:

- Limit output
- Control structure

Write:

- What constraints would help?

📘 L7_09 — Prompt Chaining

Key Ideas:

- Output → next input
- Multi-step workflows

Write:

- What process can you chain?

📘 L7_10 — Context + Memory

Key Ideas:

- Context improves output
- Too much = confusion

Write:

- What context matters most?

📘 L7_11 — Multi-Step Systems

Key Ideas:

- Combine techniques
- Full workflows

Write:

- What system could you build?

📘 L7_12 — Prompt Optimization

Key Ideas:

- Test → refine → improve
- Small changes = big results

Write:

- What prompt can you optimize?

🛠️ PART 2 — PRACTICE LAB (MOST IMPORTANT)

🔹 Exercise 1 — Prompt Upgrade

Start:

“Write about AI”

Improve:

Write a clear, structured article about AI for beginners.
Include:
- Simple explanation
- Real-world example
- 3 key takeaways

👉 Compare outputs

🔹 Exercise 2 — Role Prompting

Try:

Act as a business consultant.
Explain how AI can improve productivity.

Then change role → compare

🔹 Exercise 3 — Step Prompt

Explain AI step-by-step:
1. What it is
2. How it works
3. Example

🔹 Exercise 4 — Constraint Prompt

Explain AI in under 100 words using simple language.

🔹 Exercise 5 — Prompt Chain

1. Generate ideas

2. Expand best idea
3. Turn into outline
4. Write content

👉 This is **real workflow**

🧪 PART 3 — QUIZ

Q1

Prompt engineering is:

A. Coding
B. Designing instructions
C. Writing only
D. Guessing

✅ **Answer: B**

Q2

Best prompt includes:

A. One sentence
B. Instruction + context + output
C. Random text
D. No detail

✅ **Answer: B**

Q3

Iteration means:

A. Stop
B. Improve step-by-step
C. Delete
D. Restart

✅ **Answer: B**

Q4

Role prompting changes:

A. Speed
B. Perspective
C. Data
D. Code

✅ **Answer: B**

Q5

Prompt chaining is:

A. One prompt
B. Multiple connected steps
C. Random input
D. Short prompts

✅ **Answer: B**

🤔 PART 4 — REFLECTION

Reflection 1

What is the best prompt you've written so far?

Reflection 2

How much did output improve after refining?

Reflection 3

What mistakes do you keep making?

Reflection 4

How can prompting become a system for you?

🤖 PART 5 — PROMPT LIBRARY (SAVE THESE)

🔹 Universal Prompt

“Explain [topic] clearly, with examples, and summarize in steps”

🔹 Workflow Prompt

“Create a step-by-step workflow for [task] with tools and outputs”

🔹 Improvement Prompt

“Improve this output for clarity, structure, and usefulness: [text]”

🔹 Expert Prompt

“Act as an expert in [field] and provide a structured explanation of [topic]”

🧩 PART 6 — PROMPT DESIGN TEMPLATE

Goal:

Instruction:

Context:

Steps:

Constraints:

Output format:

📊 PART 7 — PROGRESS CHECK

✔ I can improve prompts
✔ I use structure
✔ I use roles
✔ I iterate outputs
✔ I chain prompts

🚫 PART 8 — COMMON MISTAKES

❌ Vague prompts
❌ No structure
❌ No iteration
❌ Not testing

🎯 FINAL TAKEAWAY (SECTION 7)

AI output is not random —
it is **designed through your prompts**